

Writing and Solving Equations in Context

MA.7.AR.2.2

I can define, write, solve and interpret — and decide which way a decimal answer rounds.

The last equations lesson.

Two new things: fractional coefficients, and answers that have to be whole.

Warm-Up

On your own. Four minutes.

Spaced retrieval

1. Last lesson: a jacket priced p is 40% off. What do you pay?
2. Last week: solve $4(x - 3) = 28$.
3. Last unit: write 0.4 as a fraction in lowest terms.
4. Prior grade: what is $12 \div (1/3)$?

Notes

I. Four rows, every time

- A. **DEFINE.** A sentence saying what the letter stands for.
- B. **WRITE.** The rate, the starting value and the total, in place.
- C. **SOLVE.** Inverse operations, both sides, one step at a time.
- D. **INTERPRET.** Answer the question in a sentence, with units.

Notes

II. Multiplying by $\frac{1}{8}$ is dividing by 8

- A. reciprocal — the fraction turned upside down
- B. **Multiplying by $\frac{1}{8}$ is the same as dividing by 8.**
- C. **So the way back is to MULTIPLY by 8 — not to divide by 8 again.**

Notes

III. For any other fraction, multiply by its reciprocal

- A. $(5/3)b = 35/3 \rightarrow b = (35/3)(3/5)$
- B. Turn the coefficient upside down and multiply both sides by it.
- C. **A decimal works too. $2/5 = 0.4$, and $5.6 \div 0.4 = 14$, the same answer.**

Notes

IV. What the number counts decides which way it rounds

- A. **Solve first. Then ask what the number counts.**
- B. People, buses, batches and shirts come in whole numbers. Hours, pounds and dollars do not.
- C. **A maximum rounds DOWN. A minimum rounds UP. Decide from the situation, never from the decimal.**

Example 1

A fractional coefficient.

$$(1/4)\mathbf{x} = 9$$

Before you solve: will x be bigger or smaller than 9?

Example 1

Multiply by 4.

$$(1/4)x = 9$$

$$(1/4)x \times 4 = 9 \times 4 \quad \text{multiplication property of equality}$$

$$x = 36$$

$$\text{Check: } (1/4)(36) = 9 \quad \checkmark$$

Example 1

And when the fraction is improper.

$$(5/2)x = 20$$

multiply both sides by $2/5$, the reciprocal

$$x = 8$$

$$\text{Check: } (5/2)(8) = 20 \quad \checkmark$$

Example 2

A gift card.

A \$60 gift card. Five equal purchases are made. \$12.50 is left.

How much was each purchase?

Define your letter, then write the equation. Do not solve it.

Example 2

Solve and interpret.

$$60 - 5p = 12.50$$

$$-5p = -47.50$$

$$p = 9.50$$

$$\text{Check: } 60 - 5(9.50) = 60 - 47.50 = 12.50 \quad \checkmark$$

Each purchase was \$9.50.

Example 2

Does it need rounding?

No. Dollars come in fractions of a dollar, and \$9.50 is a real price.

Example 3

Two answers, both 6.4.

A budget allows 6.4 cases of juice.	A trip needs 6.4 buses.
How many cases can you buy?	How many buses do you book?

Same number. Do they round the same way?

Example 3

No.

6 cases	7 buses
7 cases would cost more than the budget	6 buses would leave people behind
round DOWN	round UP

Example 3

And sometimes neither.

An equation says a tank has been draining for -3 hours.

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{1}{4}x = 9$$

Solve it, or answer it.

Read what the number counts before you round anything.

Whiteboards • Question 1 of 9

Answer.

$$\frac{1}{4}x = 9$$

multiply both sides by 4

Answer: $x = 36$

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{2}{3}x = 10$$

Solve it, or answer it.

Read what the number counts before you round anything.

Whiteboards · Question 2 of 9

Answer.

$$\frac{2}{3}x = 10$$

multiply by the reciprocal, $\frac{3}{2}$

Answer: $x = 15$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{7}{3}x = 21$$

Solve it, or answer it.

Read what the number counts before you round anything.

Whiteboards · Question 3 of 9

Answer.

$$\frac{7}{3}x = 21$$

multiply by the reciprocal, $\frac{3}{7}$

Answer: $x = 9$

Whiteboards • Question 4 of 9

Boards up on three.

$$\frac{2}{3}x = 10.5$$

Solve it, or answer it.

Read what the number counts before you round anything.

Whiteboards · Question 4 of 9

Answer.

$$\frac{2}{3}x = 10.5$$

multiply by 3/2: $10.5 \times 3 \div 2$

Answer: $x = 15.75$

Whiteboards • Question 5 of 9

Boards up on three.

A budget allows 15.75 students on a trip. How many go?

Solve it, or answer it.

Read what the number counts before you round anything.

Whiteboards • Question 5 of 9

Answer.

A budget allows 15.75 students on a trip. How many go?

16 would go over budget

Answer: 15 students

Whiteboards • Question 6 of 9

Boards up on three.

A trip needs 15.75 vans. How many do you book?

Solve it, or answer it.

Read what the number counts before you round anything.

Whiteboards • Question 6 of 9

Answer.

A trip needs 15.75 vans. How many do you book?

15 would leave people behind

Answer: 16 vans

Whiteboards • Question 7 of 9

Boards up on three.

Peyton has a \$40 gift card. She buys the same thing four times and has \$7.08 left. What did each one cost?

Write the equation and solve it.

Read what the number counts before you round anything.

Whiteboards • Question 7 of 9

Answer.

Peyton has a \$40 gift card. She buys the same thing four times and has \$7.08 left. What did each one cost?

$$40 - 4c = 7.08, \text{ so } 4c = 32.92$$

Answer: \$8.23 each

Whiteboards • Question 8 of 9

Boards up on three.

Shirley gives each of her eight grandchildren a \$15 gift card AND the same amount of cash.
She spends \$280 in total. How much cash each?

Write the equation and solve it.

Read what the number counts before you round anything.

Whiteboards • Question 8 of 9

Answer.

Shirley gives each of her eight grandchildren a \$15 gift card AND the same amount of cash.
She spends \$280 in total. How much cash each?

$$8(15 + c) = 280, \text{ so } 15 + c = 35$$

Answer: \$20 each

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Songs cost \$1.75 each. Georgina has a \$50 gift card. How many songs can she buy, and how much is left?

Write your answer in sentences.

This one is written work. Say why you rounded the way you did.

Whiteboards • Question 9 of 9

Answer.

Songs cost \$1.75 each. Georgina has a \$50 gift card. How many songs can she buy, and how much is left?

$$50 \div 1.75 = 28.57\dots; \text{ 29 songs cost } \$50.75$$

Answer: 28 songs, \$1.00 left

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.